

JIACHENG LIANG

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RESEARCH INTEREST

My research focuses on **large language models** and **trustworthy AI**, spanning LLM post-training and alignment, retrieval-augmented generation, agentic LLM systems, and robust model deployment. I build end-to-end LLM pipelines for improving the reliability, safety, and real-world effectiveness of AI systems across applications such as recommendation, LLM serving, and AI security.

EDUCATION

Stony Brook University & Penn State University

Ph.D. Candidate in Computer Science, GPA: 4.0/4.0

Stony Brook, NY

Sep 2022 – Dec 2026 (Expected)

University of Electronic Science and Technology of China

B.Eng. in Software Engineering – International Elite Class

Chengdu, China

Sep 2018 – Jun 2022

EXPERIENCE

Prime Video & Amazon MGM Studios

Applied Scientist Intern

Seattle, WA

Jun 2026 – Present

- **Reinforcement Learning for Personalized Recommendation.** Exploring RL-based methods for personalized recommendation, with a focus on generating multi-intent retrieval queries that balance long-term user preferences, session context, and time-sensitive trends.

Amazon AGI Foundation – Nova Responsible AI Team

Applied Scientist Intern

Boston, MA

May 2025 – Aug 2025

- **Adaptive Red-Teaming for RLHF.** Identified systemic weaknesses in RLHF pipelines where the policy and reward model fail in tandem; proposed an adaptive red-teaming + end-to-end repair framework for more robust alignment. (*Pub. ARES, ACL '26*)

PUBLICATIONS

Advisor: Prof. Ting Wang (ALPS-Lab). Entries are first / co-first authored unless tagged **co-author**.

Attack & Red-Teaming

AutoRAN: Automated Hijacking of Safety Reasoning in Large Reasoning Models

ACL '26 

Automatically hijacks internal safety reasoning in LLMs — turning chain-of-thought transparency into an attack surface.

ARES: Adaptive Red-Teaming and End-to-End Repair of Policy-Reward Systems

Amazon AGI · ACL '26 

Exposes joint failures of policy & reward models in RLHF; proposes adaptive red-teaming + end-to-end repair.

GraphRAG under Fire

IEEE S&P '26 (Oakland) 

First poisoning attack on GraphRAG — corrupts entity-relation graphs to hijack downstream knowledge retrieval.

Defense & Alignment

Data to Defense: The Role of Curation in Customizing LLMs Against Jailbreaking Attacks

EMNLP '25 

End-to-end curation across LLM customization stages — achieves 100% safe-response rate against jailbreaks.

Robust Semantic Cache for LLM Serving

Under review

Indexes cache on request + speculatively decoded tokens — provably resists cache poisoning while improving retrieval relevance.

RASA: Routing-Aware Safety Alignment for Mixture-of-Experts Models

Under review 

Routing-aware bi-level alignment that robustifies both experts and routers in MoE LLMs against jailbreaks.

MAGE: Safeguarding LLM Agents against Long-Horizon Threats via Shadow Memory

Preprint **co-author** 

Agentic shadow memory distills safety-critical context to detect and mitigate long-horizon threats before unsafe actions execute.

Dynamic Token Reweighting for Robust Vision-Language Models

CVPR '26 **co-author** 

First KV-cache reweighting method for multimodal jailbreak defense — robustifies VLMs without retraining.

RobustKV: Defending Large Language Models against Jailbreak Attacks via KV Eviction

ICLR '25 **co-author** 

Selectively evicts low-importance KV-cache tokens — neutralizes jailbreaks while preserving benign performance.

Your Agent Can Defend Itself Against Backdoor Attacks

Preprint **co-author** 

Self-defense for LLM agents by enforcing consistency between planning, execution, and user instructions.

Benchmark & Evaluation

AgentLAB: Benchmarking LLM Agents Against Long-Horizon Attacks

ICML '26 **co-author** 

First extensible benchmark for long-horizon attacks on LLM agents — uncovers vulnerabilities beyond single-turn defenses.

Model Extraction Attacks Revisited (MEBench)

AsiaCCS '24 

Open-source platform assessing model-extraction vulnerabilities across real MLaaS APIs — unifies attacks, metrics, and victim models.

WaterPark: A Robustness Assessment of Language Model Watermarking

EMNLP '25 

Systematization of LLM watermarking — standardized robustness eval, novel attack, and corresponding defenses.

Reasoning or Retrieval? A Study of Answer Attribution on Large Reasoning Models

ICLR '26 **co-author** 

Diagnostic study attributing LRM answers between genuine reasoning and parametric retrieval.